Database Technical Documentation

Algérie Poste E-Commerce Platform



Complete Database Architecture,
Entity Relationships, and Normalization Analysis

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Chapter 1

Introduction

1.1 Purpose

This document provides comprehensive technical documentation for the Algérie Poste E-Commerce database. It is intended for:

- Backend developers implementing API features
- Database administrators managing the production environment
- New team members onboarding to the project
- Auditors reviewing data architecture

1.2 Scope

The database supports a multi-vendor e-commerce marketplace with the following key features:

1. **Multi-vendor Marketplace** – Multiple sellers with individual storefronts
2. **Escrow Payment System** – Secure buyer protection with fund holding
3. **Hybrid Tracking** – Algérie Poste API + internal last-mile tracking
4. **Home Delivery** – Courier management with GPS tracking
5. **OTP Verification** – Secure package handoff confirmation

1.3 Technology Stack

1.4 Document Conventions

- `PK` – Primary Key
- `FK` – Foreign Key
- `UK` – Unique Key

Component	**Technology**
Database Engine	MySQL 8.0+
Character Set	utf8mb4 (Arabic/French support)
Storage Engine	InnoDB (ACID transactions)
ORM	TypeORM (NestJS backend)

Table 1.1: Technology Stack

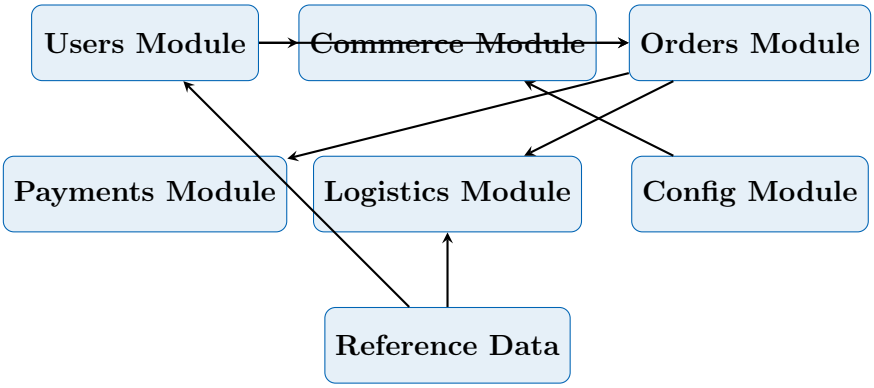
- `1:N` – One-to-Many relationship
- `1:1` – One-to-One relationship
- `N:M` – Many-to-Many relationship

Chapter 2

Database Architecture

2.1 Overview

The database is organized into logical modules:



2.2 Module Descriptions

2.2.1 Users Module

Handles user accounts, authentication, addresses, and seller verification.

- `users` – Central user accounts
- `user_addresses` – Shipping/billing addresses
- `seller_registrations` – Seller verification documents

2.2.2 Commerce Module

Manages stores, products, categories, and reviews.

- `stores` – Seller storefronts
- `products` – Product catalog
- `categories` – Hierarchical categories
- `reviews` – Product reviews

2.2.3 Orders Module

Handles order lifecycle from cart to delivery.

- `orders` – Customer orders
- `order_items` – Line items with snapshots

2.2.4 Payments Module

Manages payment transactions and escrow accounts.

- `payments` – Payment transactions
- `escrow_accounts` – Fund holding
- `disputes` – Payment disputes

2.2.5 Logistics Module

Handles shipping with hybrid tracking (Algérie Poste + last-mile).

- `shipments` – Shipping records
- `tracking_events` – Tracking history
- `couriers` – Delivery personnel
- `courier_assignments` – Delivery assignments
- `delivery_zones` – Geographic zones
- `delivery_ratings` – Courier ratings

2.2.6 Reference Data

Static reference tables.

- `wilayas` – 58 Algerian provinces
- `themes` – Store themes
- `plans` – Subscription plans

Chapter 3

Entity-Relationship Model

3.1 Core Entities

3.1.1 Users Entity

The `users` table is the central entity for all user types.

Attribute	Type	Description
id	INT PK	Auto-increment primary key
email	VARCHAR(255) UK	Unique login email
password_hash	VARCHAR(255)	Bcrypt hashed password
full_name	VARCHAR(255)	Display name
phone	VARCHAR(20)	Phone (+213...)
account_type	ENUM	buyer, seller, both
account_status	ENUM	active, inactive, pending, suspended
language_preference	VARCHAR(10)	fr, ar, en

Table 3.1: Users Entity Attributes

3.1.2 Products Entity

Attribute	Type	Description
id	INT PK	Auto-increment primary key
store_id	INT FK	Reference to stores
category_id	INT FK	Reference to categories
name	VARCHAR(255)	Product title
price	DECIMAL(10,2)	Price in DZD
stock	INT	Available quantity
status	ENUM	active, draft, sold_out
images	JSON	Array of image URLs

Table 3.2: Products Entity Attributes

3.1.3 Orders Entity

Attribute	Type	Description
id	INT PK	Auto-increment primary key
buyer_id	INT FK	Customer who placed order
store_id	INT FK	Seller's store
order_number	VARCHAR(50) UK	Human-readable ID
status	ENUM	pending, confirmed, shipped, delivered, etc.
total_amount	DECIMAL(10,2)	Total in DZD
shipping_address	JSON	Delivery address snapshot

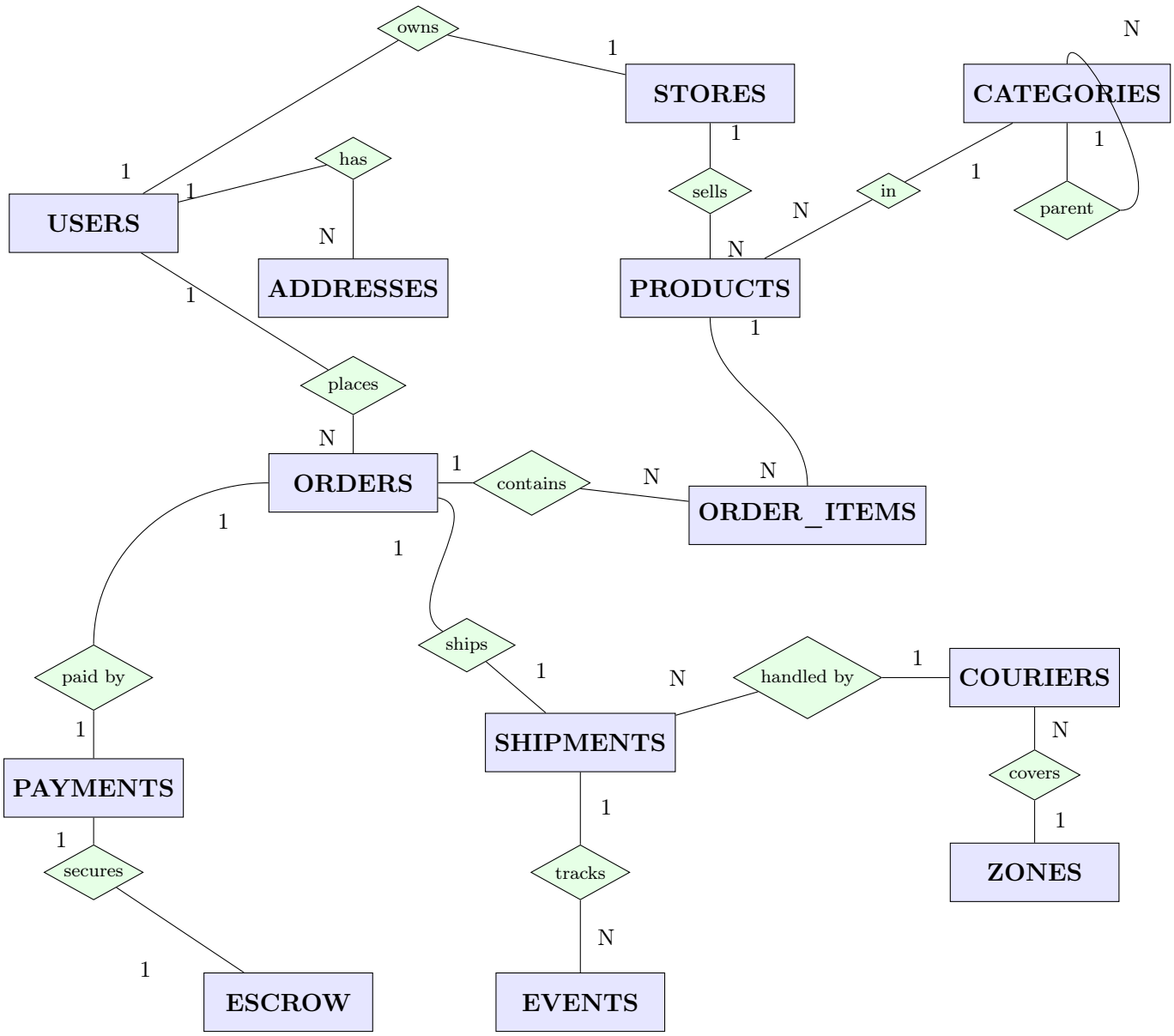
Table 3.3: Orders Entity Attributes

3.1.4 Shipments Entity (Enhanced)

Attribute	Type	Description
id	INT PK	Auto-increment primary key
order_id	INT FK	Related order
tracking_number	VARCHAR(100) UK	Algérie Poste tracking
last_mile_tracking	VARCHAR(100) UK	Internal tracking (AP-LM-xxx)
status	ENUM	pending, at_bureau, out_for_delivery, delivered
delivery_type	ENUM	bureau_pickup, home_delivery
courier_id	INT FK	Assigned courier
current_latitude	DECIMAL(10,7)	GPS latitude
current_longitude	DECIMAL(10,7)	GPS longitude
delivery_otp	VARCHAR(6)	6-digit verification code
delivery_attempts	INT	Number of attempts (max 3)

Table 3.4: Shipments Entity Attributes (Enhanced for Last-Mile)

3.2 Relationship Diagram



3.3 Cardinality Matrix

Relationship	**Type**	**Description**
User → Addresses	1:N	One user has many addresses
User → Store	1:1	One seller owns one store
User → Orders	1:N	One buyer places many orders
Store → Products	1:N	One store sells many products
Category → Products	1:N	One category contains many products
Category → Subcategories	1:N	Self-referencing hierarchy
Order → Items	1:N	One order has many line items
Order → Payment	1:1	One order has one payment
Order → Escrow	1:1	One order has one escrow account
Order → Shipment	1:1	One order has one shipment
Shipment → Events	1:N	One shipment has many tracking events
Shipment → Courier	N:1	Many shipments assigned to one courier
Courier → Zone	N:1	Many couriers cover one zone

Table 3.5: Relationship Cardinality Matrix

Chapter 4

Normalization Analysis

4.1 Normal Forms Applied

The database schema adheres to the following normal forms:

4.1.1 First Normal Form (1NF)

Requirements:

- All tables have a primary key
- All columns contain atomic values
- No repeating groups

Status: All tables satisfy 1NF, with documented exceptions for JSON fields used for flexibility (images, attributes, addresses).

4.1.2 Second Normal Form (2NF)

Requirements:

- Must satisfy 1NF
- No partial dependencies on composite keys

Status: All tables use single-column primary keys, eliminating partial dependencies.

4.1.3 Third Normal Form (3NF)

Requirements:

- Must satisfy 2NF
- No transitive dependencies

Status: Most tables satisfy 3NF. Intentional denormalizations are documented.

4.2 Normalization Status by Table

Table	1NF	2NF	3NF	Notes
users				Fully normalized
user_addresses				Fully normalized
seller_registrations				Fully normalized
stores				Fully normalized
products	△		△	JSON fields, derived rating
orders	△		△	JSON address snapshots
order_items	△		△	Product snapshots
payments				Fully normalized
escrow_accounts				**Critical: Fully normalized**
shipments			△	Courier info cached
tracking_events				Fully normalized
couriers				Fully normalized
wilayas				Reference data

Table 4.1: Normalization Status (= Pass, △ = Intentional deviation)

4.3 Intentional Denormalizations

4.3.1 Product Snapshots in Order Items

Fields: `product_name`, `product_price`, `product_snapshot`

Justification:

- Products may be renamed or deleted after order
- Prices may change
- Legal requirement to preserve original transaction details
- Common e-commerce pattern (Amazon, Shopify)

4.3.2 Address Snapshots in Orders

Fields: `shipping_address` (JSON), `billing_address` (JSON)

Justification:

- User may modify or delete addresses after order
- Historical orders must show original delivery location
- Immutability requirement for audit trails

4.3.3 Derived Fields in Products

Fields: `rating`, `review_count`

Calculation: rating = AVG(reviews.rating)
review_count = COUNT(reviews)

Justification:

- Avoids expensive JOIN on every product listing
- Updated via application logic when reviews change

Chapter 5

Data Dictionary

5.1 Users Table

```sql
CREATE TABLE users (
  id INT AUTO_INCREMENT PRIMARY KEY,
  email VARCHAR(255) NOT NULL UNIQUE,
  password_hash VARCHAR(255) NOT NULL,
  full_name VARCHAR(255),
  phone VARCHAR(20),
  bio TEXT,
  account_type ENUM('buyer', 'seller', 'both')
    DEFAULT 'buyer',
  account_status ENUM('active', 'inactive',
    'pending', 'suspended') DEFAULT 'pending',
  language_preference VARCHAR(10) DEFAULT 'fr',
  last_login TIMESTAMP NULL,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
    ON UPDATE CURRENT_TIMESTAMP
);
```

Listing 5.1: Users Table DDL

5.2 Products Table

```sql
CREATE TABLE products (
  id INT AUTO_INCREMENT PRIMARY KEY,
  store_id INT NOT NULL,
  category_id INT,
  name VARCHAR(255) NOT NULL,
  slug VARCHAR(255) NOT NULL UNIQUE,
  description TEXT,
  price DECIMAL(10,2) NOT NULL,
  compare_price DECIMAL(10,2),
  stock INT DEFAULT 0,
  sku VARCHAR(100),
```

```
  status ENUM('active', 'draft', 'sold_out')
    DEFAULT 'draft',
  images JSON,
  attributes JSON,
  weight DECIMAL(5,2),
  dimensions JSON,

  FOREIGN KEY (store_id) REFERENCES stores(id),
  FOREIGN KEY (category_id) REFERENCES categories(id)
);
```

Listing 5.2: Products Table DDL

5.3 Shipments Table (Enhanced)

```
CREATE TABLE shipments (
  id INT AUTO_INCREMENT PRIMARY KEY,
  order_id INT NOT NULL UNIQUE,
  tracking_number VARCHAR(100) NOT NULL UNIQUE,
  last_mile_tracking VARCHAR(100) UNIQUE,
  provider ENUM('ems_standard', 'ems_express',
    'colis') NOT NULL,
  status ENUM('pending', 'label_created',
    'picked_up', 'in_transit', 'at_bureau',
    'out_for_delivery', 'delivery_attempted',
    'delivered', 'failed_delivery', 'returned')
    DEFAULT 'pending',
  delivery_type ENUM('bureau_pickup',
    'home_delivery', 'relay_point')
    DEFAULT 'home_delivery',
  courier_id INT,
  current_latitude DECIMAL(10,7),
  current_longitude DECIMAL(10,7),
  delivery_otp VARCHAR(6),
  otp_expires_at TIMESTAMP,
  delivery_attempts INT DEFAULT 0,
  max_delivery_attempts INT DEFAULT 3,

  FOREIGN KEY (order_id) REFERENCES orders(id),
  FOREIGN KEY (courier_id) REFERENCES couriers(id)
);
```

Listing 5.3: Shipments Table DDL with Last-Mile Tracking

5.4 Escrow Accounts Table

```
CREATE TABLE escrow_accounts (
  id INT AUTO_INCREMENT PRIMARY KEY,
  order_id INT NOT NULL UNIQUE,
```

```
  payment_id INT NOT NULL,
  amount DECIMAL(10,2) NOT NULL,
  status ENUM('pending', 'locked', 'released',
    'disputed', 'refunded') DEFAULT 'pending',
  locked_at TIMESTAMP,
  releasable_at TIMESTAMP,
  released_at TIMESTAMP,

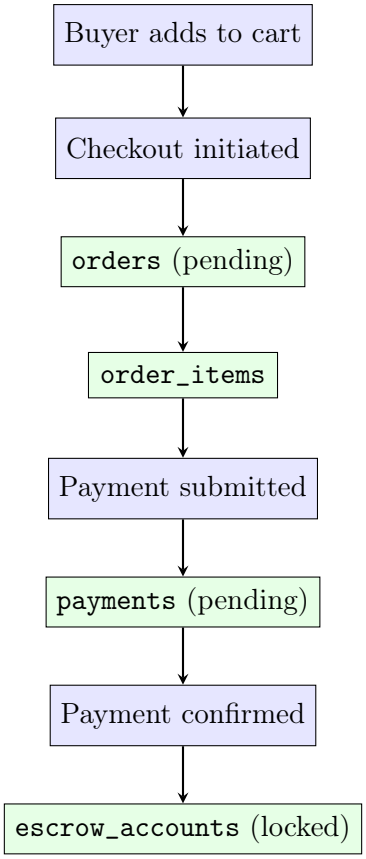
  FOREIGN KEY (order_id) REFERENCES orders(id),
  FOREIGN KEY (payment_id) REFERENCES payments(id)
);
```

Listing 5.4: Escrow Accounts Table DDL

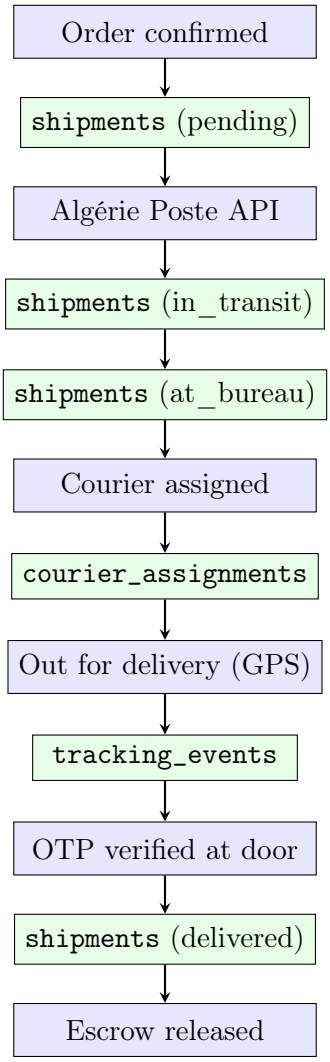
Chapter 6

Data Flows

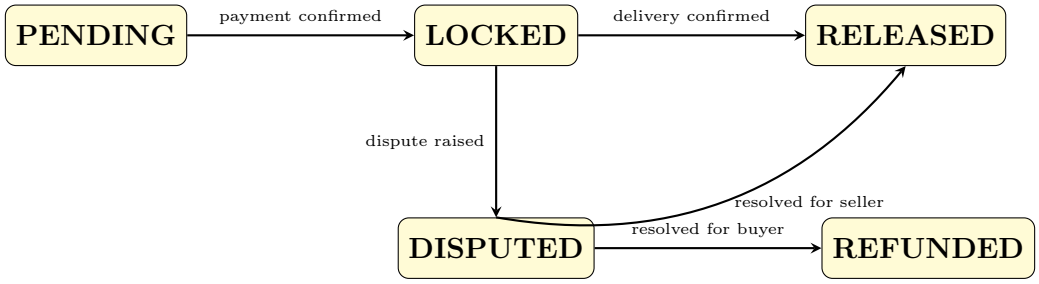
6.1 Order Creation Flow



6.2 Shipping Flow



6.3 Escrow Flow



Chapter 7

Indexes and Performance

7.1 Index Strategy

7.1.1 Primary Indexes

All tables use auto-increment integer primary keys for optimal JOIN performance.

7.1.2 Unique Indexes

- `users.email` – Fast login lookup
- `products.slug` – URL routing
- `orders.order_number` – Customer-facing lookup
- `shipments.tracking_number` – Tracking queries

7.1.3 Foreign Key Indexes

All foreign keys are automatically indexed by MySQL InnoDB.

7.1.4 Composite Indexes

```
-- Products by store and status
CREATE INDEX idx_products_store_status
  ON products(store_id, status);

-- Orders by buyer and status
CREATE INDEX idx_orders_buyer_status
  ON orders(buyer_id, status);

-- Shipments awaiting delivery
CREATE INDEX idx_shipments_delivery_pending
  ON shipments(status, delivery_type);

-- Tracking events by shipment and time
CREATE INDEX idx_tracking_shipment_time
  ON tracking_events(shipment_id, event_time DESC);
```

Listing 7.1: Key Composite Indexes

7.2 Query Optimization Tips

1. Use `EXPLAIN` to analyze query plans
2. Avoid `SELECT *` – specify needed columns
3. Use pagination with `LIMIT` and `OFFSET`
4. Cache derived data (product ratings)
5. Use connection pooling (TypeORM default)

Chapter 8

Security Considerations

8.1 Data Protection

8.1.1 Password Storage

Passwords are hashed using bcrypt with salt rounds of 12:

```
import * as bcrypt from 'bcrypt';

const saltRounds = 12;
const hash = await bcrypt.hash(password, saltRounds);
```

Listing 8.1: Password Hashing Example

8.1.2 Sensitive Data

- `password_hash` – Never exposed via API
- `ssid`, `cnrc_number` – Seller documents, admin-only access
- `ccp_account` – Payment info, encrypted at rest

8.2 Access Control

Row-level security is implemented at the application layer:

- Buyers can only view their own orders
- Sellers can only view their store's orders
- Couriers can only view assigned shipments
- Admins have full access with audit logging

8.3 Audit Trail

The `audit_logs` table records all sensitive operations:

```
audit_logs (
  user_id     -- Who performed the action
  action      -- CREATE, UPDATE, DELETE
  entity_type -- 'order', 'payment', etc.
  entity_id   -- Affected record ID
  old_values  -- JSON of previous state
  new_values  -- JSON of new state
  ip_address  -- Request IP
  user_agent  -- Browser/client info
  created_at  -- Timestamp
)
```

Listing 8.2: Audit Log Structure

Chapter 9

Backup and Recovery

9.1 Backup Strategy

9.1.1 Full Backup

```
mysqldump -u root -p \
  --single-transaction \
  --routines \
  --triggers \
  ap_ecommerce > backup_$(date +%Y%m%d_%H%M%S).sql
```

Listing 9.1: Full Database Backup

9.1.2 Incremental Backup

Enable binary logging for point-in-time recovery:

```
[mysqld]
log-bin=mysql-bin
binlog_format=ROW
expire_logs_days=7
```

Listing 9.2: MySQL Binary Log Configuration

9.2 Recovery Procedures

9.2.1 Full Restore

```
mysql -u root -p ap_ecommerce < backup_20251206.sql
```

Listing 9.3: Database Restore

9.2.2 Point-in-Time Recovery

```
mysqlbinlog mysql-bin.000001 \
  --stop-datetime="2025-12-06 14:00:00" \
  | mysql -u root -p ap_ecommerce
```

Listing 9.4: Point-in-Time Recovery

Appendix A

Complete Table List

Table	Module	Description
users	Users	User accounts and authentication
user_addresses	Users	Shipping/billing addresses
seller_registrations	Users	Seller verification documents
stores	Commerce	Seller storefronts
themes	Commerce	Store visual themes
categories	Commerce	Product category hierarchy
products	Commerce	Product catalog
reviews	Commerce	Product reviews and ratings
orders	Orders	Customer orders
order_items	Orders	Order line items
payments	Payments	Payment transactions
escrow_accounts	Payments	Held funds for buyer protection
disputes	Payments	Payment/delivery disputes
shipments	Logistics	Shipping with GPS tracking
tracking_events	Logistics	Shipment tracking history
couriers	Logistics	Delivery personnel
courier_assignments	Logistics	Courier-shipment assignments
delivery_zones	Logistics	Geographic delivery areas
delivery_ratings	Logistics	Courier ratings
wilayas	Reference	58 Algerian provinces
plans	Config	Subscription plans
subscriptions	Config	Store subscriptions
notifications	Config	User notifications
audit_logs	Config	System audit trail

Table A.1: Complete Table Reference

Appendix B

Environment Variables

```
# Database Connection
DB_HOST=localhost
DB_PORT=3306
DB_USERNAME=root
DB_PASSWORD=your_secure_password
DB_DATABASE=ap_ecommerce

# TypeORM Settings
DB_SYNCHRONIZE=false  # Never true in production!
DB_LOGGING=false
DB_SSL=true           # Enable for production

# Connection Pool
DB_POOL_SIZE=10
DB_CONNECTION_TIMEOUT=30000
```

Listing B.1: Environment Configuration